OPEN SOURCE · AI AGENT OPERATIONS

Muster

An AI engineering team you run from your phone. **Grokbot is the control layer.** Claude Code, Pi and OpenCode agents on five model families design, build, review and ship, and every step is recorded.

~800
worker runs in October 2026

1,508
logged model-routing decisions

5
model families at work

3
agent harnesses on one bus

THE PROBLEM

One AI agent is a conversation. Shipping software takes a team.

- Real work needs design, building, independent review, integration and release, every time.
- A single agent forgets, stalls silently and grades its own work.
- One model or one vendor is a single point of failure and of cost.
- The person in charge ends up babysitting terminals instead of deciding outcomes.

WHO IT IS FOR

Builders who want an AI team, not another chat window

Solo founders and small studios

Run several products at once with a team of agents that works while you are away.

Engineering leads

Put AI agents behind the same process as people: briefs, review by another model, proof before merge.

AI platform teams

Mix harnesses and vendors, route by cost and quota, and fail over when a provider walls.

GROKBOT: THE CONTROL LAYER

You talk to Grokbot. Grokbot runs the team.

One front door, chat or voice

The human gives goals and answers to Grokbot on the phone. Grokbot relays the exact words to the architect and orchestrator seats and wakes them.

Live board, audited tools

Grokbot reads lanes, asks, goals and releases through the Muster gateway: about 60 typed tools, bearer roles, an audit log, and no shell tool.

Decisions as one card

Every question reaches the human prepared, with options and a recommendation. One tap answers it and the queue closes it with evidence.

Live intelligence from X

Grokbot reads X for model launches, prices and terms. Signals land in the research desk, and a free model that passes a probe is offered with one tap.

Independent reviewer

Grok is reserved to review risky fixes: a different model family from the author checks the impact map and the tests before merge.

Guard rails stay human

Spending, messages to real people and production data go out only on the human's own tap. Grokbot prepares, the human decides.

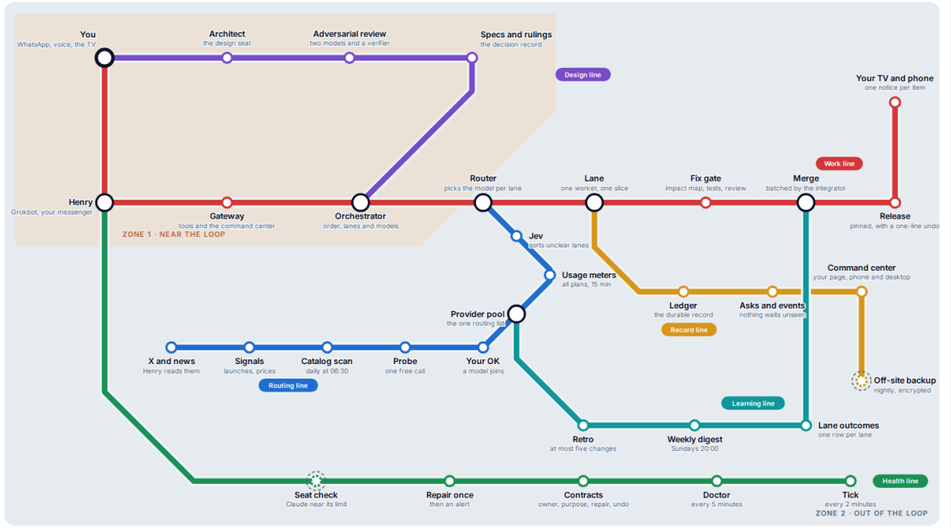
HOW IT WORKS

The system at a glance

MUSTER · SYSTEM MAP

One system, six lines

Each coloured line is a flow through muster. An interchange is a part that two flows share. Zone 1 runs while you watch, near the loop. Zone 2 runs while you are away, out of the loop. Tap a line or a station.



Zone 1: near the loop, you are watching Zone 2: out of the loop, it runs on its own Being built now

The lines

- **Work line** — One piece of work, from your message to your TV
- **Design line** — Plans are attacked before anyone builds them
- **Routing line** — Which model does the job, and how new models get in
- **Health line** — Caretakers check every part, around the clock
- **Record line** — Everything is written down, so nothing is lost
- **Learning line** — Every week the numbers change one owner

Tap a line again to show all lines.

You

Work line · Design line

the operator. Every line starts or ends with you. You give goals in plain words, by message or voice, and only you approve spending, anything public, production data and logins. A seat moves only when you type the phrase.

Zone 1: near the loop.

THE PIECES

A bus that turns agents into an accountable team

Bus and ledger

send, ask, report, wait. A daemon on a local socket with SQLite storage and a spool for restarts.

Seats

The architect turns goals into reviewed blueprints. The orchestrator writes briefs, routes, verifies and merges.

Lanes

One worker, one written brief, its own folder. Fast, cheap models by default.

Model router

Role chains per lane class, quota meters per provider, wall detection, fallbacks, and a logged reason for every placement.

Command center

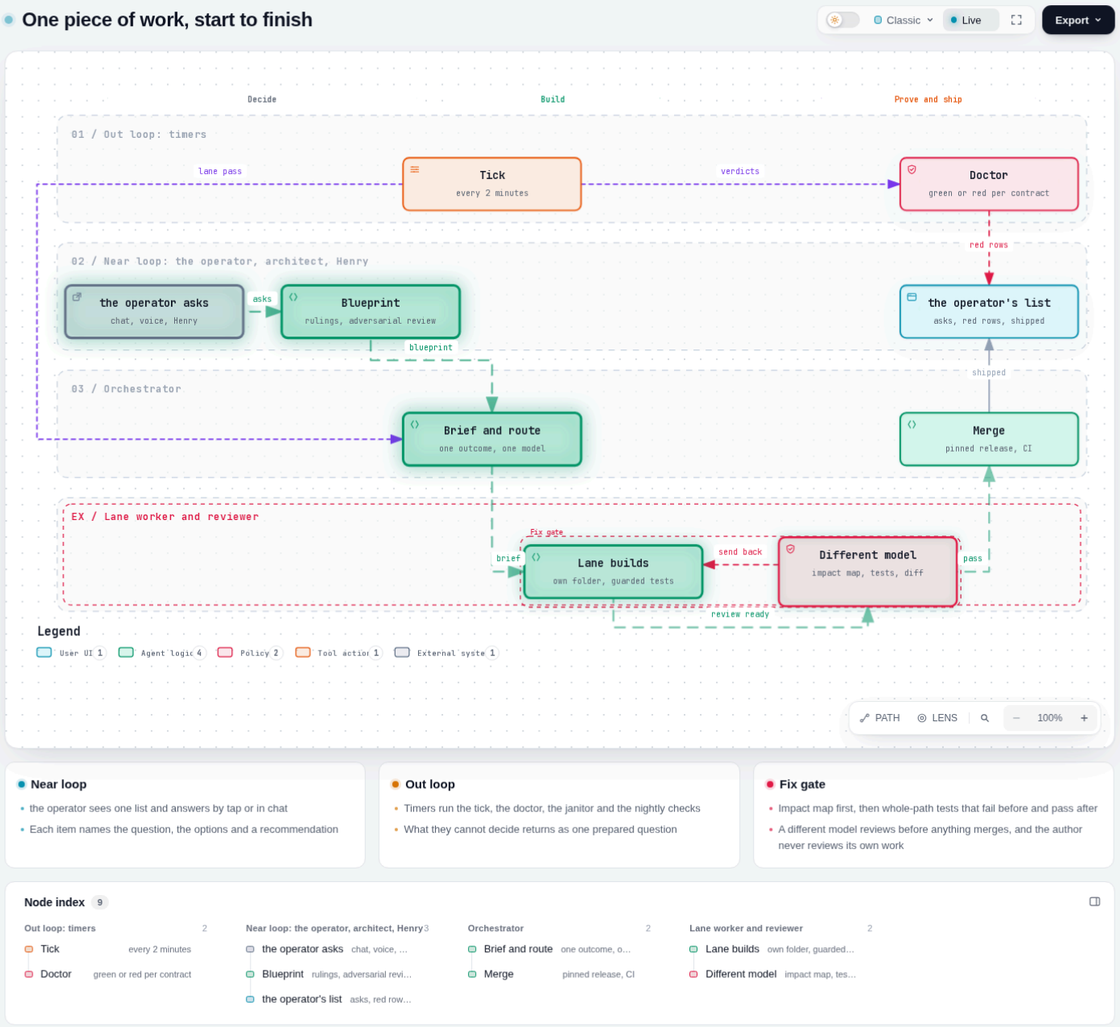
One live page for phone, desktop and TV: goals, lanes with their model, what needs you, what shipped today.

Out loop

No model needed: a tick every 2 minutes, a doctor that turns each broken contract red, a janitor.

FROM A PHONE MESSAGE TO A MERGED RELEASE

One piece of work, end to end

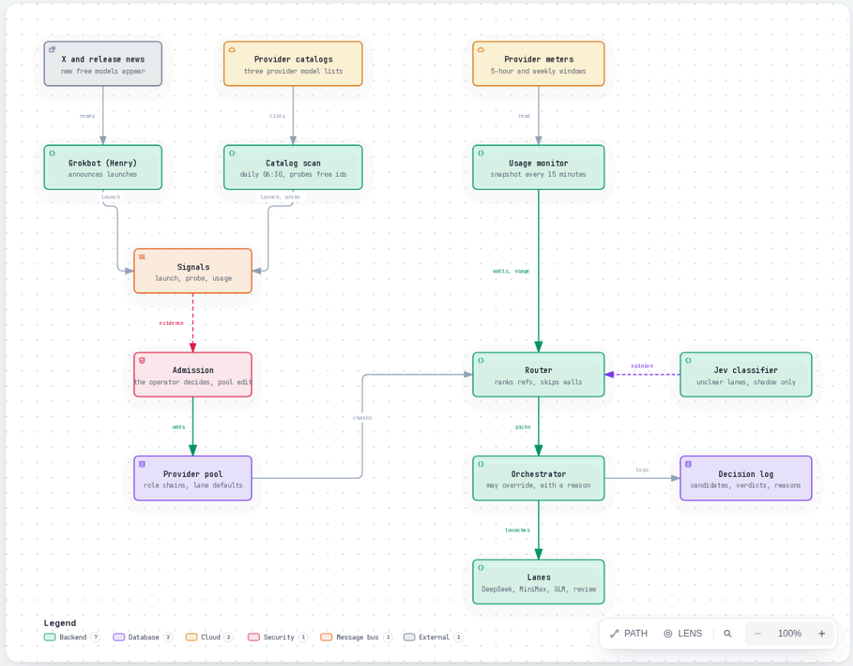


CHEAPEST MODEL THAT FITS, SKIPPING QUOTA WALLS

The model router

Model routing and new-model intake

Classic · Export



Quota-aware

- Each provider's own meter is read on a timer
- A wall skips that provider until its reset, no lane is lost
- MiniMax refills every 5 hours, so the router keeps work on it

New models come in through evidence

- Grokbot reads X and the daily scan reads the catalogs
- A free model is probed once before anyone sees it
- Admission to the pool is the operator's decision, never automatic

Every choice is logged

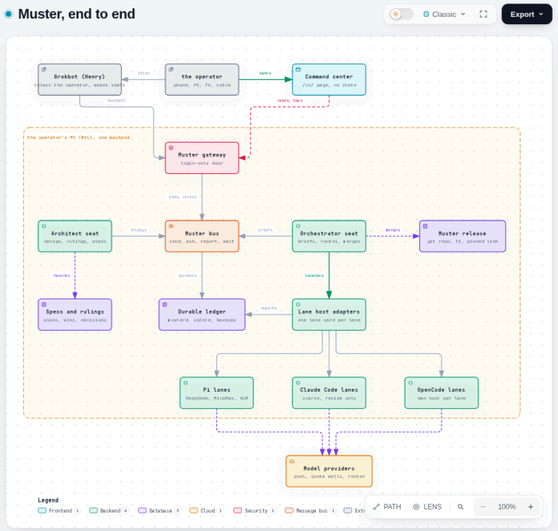
- Each lane placement keeps its candidates, verdicts and reasons
- An override by the orchestrator is recorded as one
- Jev classifies unclear lanes in shadow mode and does not decide yet

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X and release news new free …	Signals launch, probe, usage	Usage monitor snapshot every …	Decision log candidates, verdict…
Provider catalogs three provide…	Admission the operator decides, …	Router ranks refs, skips walls	Lanes DeepSeek, MiniMax, GLM,…
Grokbot (Henry) announces lau…	Provider pool role chains, lane d…	Jev classifier unclear lanes, sha…	
Catalog scan daily 06:30, probe…	Provider meters 5-hour and we…	Orchestrator may override, with …	

ARCHITECTURE

Every component and how they connect



Harness-neutral

- One bus and one lane card for Pi, Claude Code and OpenCode
- A lane is a brief, a folder and a report, whatever runs it
- A seat can move to another harness without changing the bus

Control layers

- Chain of command: the operator, then architect and Grokbot, then orchestrator, then lanes
- Fix gate: impact map, whole-path tests, independent review
- Spend, outside messages, production data and his devices stay with the operator

Durable and logged

- The ledger is one durable store with backups, owned by musterd
- Every lane model choice is logged with its candidates and reasons
- The command center keeps no state, it renders the ledger

Near loop and out loop

- Near loop: the operator's list, one page, he answers by tap or in chat
- Out loop: timers, doctor and janitor run while he is away
- Anything the out loop cannot decide becomes one prepared question

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Architecture component 4

Grokbot (Henry)	relays the operator, walk…
the operator	phone, PC, TV, voice
Command center	/cc/ page, no state
Model providers	pool, quota walls, router

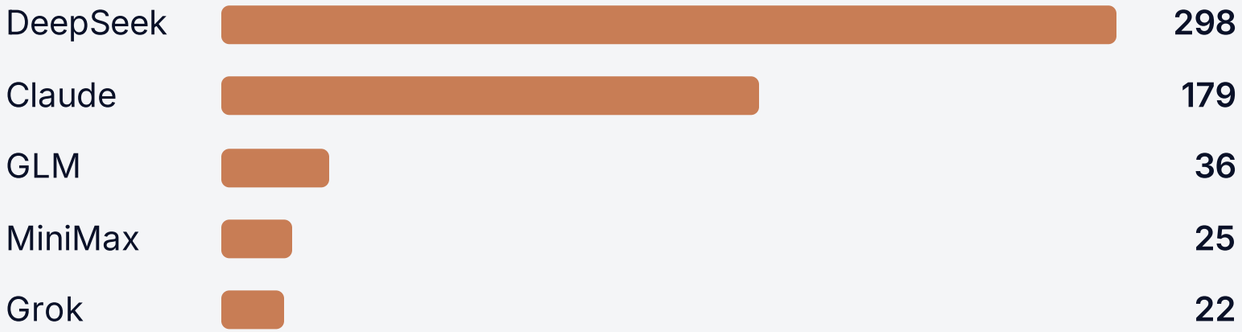
the operator's PC (WSL), one machine 11

Muster gateway	login-only door
Architect seat	design, rulings, plans
Muster bus	send, ask, report, wait
Orchestrator seat	briefs, routes, merges
Muster release	git repo, CI, pinned link
Specs and rulings	plans, wiki, decisions
Durable ledger	musterd, sqlite, backups
Lane host adapters	one lane card per lane
Pi lanes	DeepSeek, MiniMax, GLM
Claude Code lanes	scarce, review only
OpenCode lanes	own host per lane

PROOF, NOT A PROTOTYPE

In daily use on real product work since September 2026

Worker runs in October, by model family



The codebase

~94,500 lines of TypeScript

~60,000 lines of tests in **208** test files

4 packages with enforced import direction

Every fix passes a gate: an impact map, a test that fails before the fix, and a review on another model family. A status is shown only with its proof.

WHAT WE LEARNED

A change for one harness never breaks another

- The orchestrator ran for two days on a second harness. It went quiet there, and a fix made for that harness broke the first.
- The answer is structural: one wake contract for every harness, each harness a thin adapter, one conformance suite run on both before any merge.
- Reliability is designed, reviewed and tested like any feature.

NEXT

Hosting Muster on Grokbot itself

1. Package the ledger daemon, gateway and out-loop timers as one service.
2. Give each lane its own sandbox behind the existing launch interface.
3. Point Grokbot's tool calls at the local gateway. Tools, roles and audit stay the same.
4. Add Grok models as builders as well as reviewers. The router already ranks and fails over.
5. Keep the approval rules: spending, outside messages and production data stay with the human.

Grokbot becomes the team's home, not only its messenger.

TRY IT

Muster

Message Grokbot. The team builds, reviews and ships.
You decide what matters.

```
scripts/setup.sh
muster lanes
muster route pick --lane demo --task-kind research
muster seat show orchestrator
```